

Logistic Regression

- Linear regression predicts the numerical response but is not suitable for predicting the categorical variables.
- When categorical variables are involved, it is called classification problem.
- Logistic regression is suitable for binary classification problem.

Logistic Regression

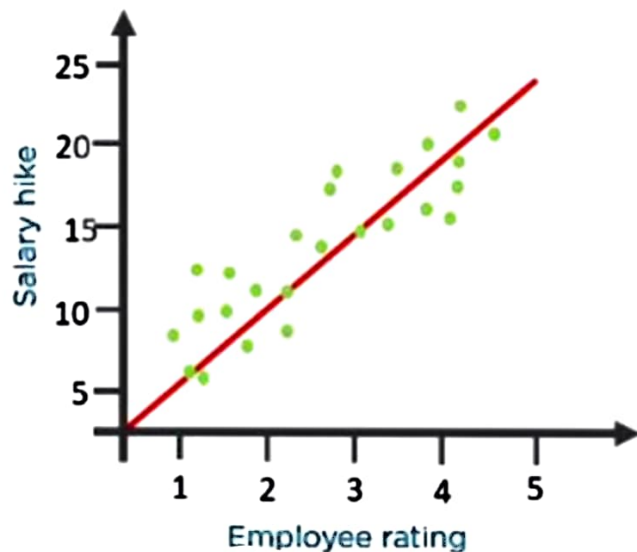
For example, the following scenarios are instances of predicting categorical variables.

1. Is the mail spam or not spam? The answer is yes or no. Thus, categorical dependent variable is a binary response of yes or no.
2. If the student should be admitted or not is based on entrance examination marks. Here, categorical variable response is admitted or not.

Logistic Regression

How Does the Logistic Regression Algorithm Work?

- Consider the following example:
- An organization wants to determine an employee's salary increase based on their performance.
- For this purpose, a linear regression algorithm will help them decide.
- Plotting a regression line by considering the employee's performance as the independent variable, and the salary increase as the dependent variable will make their task easier.



Logistic Regression

- Now, what if the organization wants to know whether an employee would get a promotion or not based on their performance?
- The above linear graph won't be suitable in this case.
- As such, we clip the line at zero and one, and convert it into a sigmoid curve (S curve).
- Based on the threshold values, the organization can decide whether an employee will get a salary increase or not.



Logistic Regression

- The dataset of pass or fail in an exam of 5 students is given in the table.
 - Use logistic regression as classifier to answer the following questions.
1. Calculate the probability of pass for the student who studied 33 hours.
 2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

Logistic Regression

- To understand logistic regression, let's go over the odds of success.

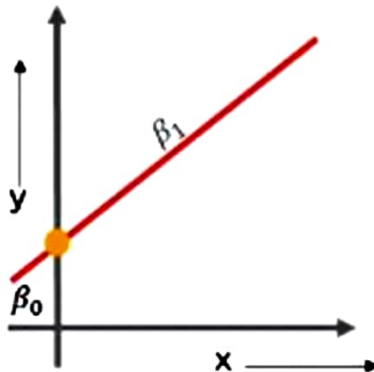
- $$\text{Odds}(\theta) = \frac{\text{Probability of an event happening}}{\text{Probability of an event not happening}}$$

- $$\text{Odds}(\theta) = \frac{p}{1-p}$$

- The values of odds range from **zero to ∞** and the values of probability lies between **zero and one**.

- Consider the equation of a straight line:

- $$y = \beta_0 + \beta_1 * x$$



Logistic Regression

- Now to predict the odds of success, we take log on odds formula:

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x$$

- Exponentiating both the sides, we have:

$$e^{\ln(x)} = x$$

$$e^{\ln\left(\frac{p(x)}{1-p(x)}\right)} = e^{\beta_0 + \beta_1 x}$$

$$\left(\frac{p(x)}{1-p(x)}\right) = e^{\beta_0 + \beta_1 x}$$

Logistic Regression

- Let $Y = e^{\beta_0 + \beta_1 * x}$
- Then $\frac{p(x)}{1 - p(x)} = Y$
- $p(x) = Y(1 - p(x))$
- $p(x) = Y - Y(p(x))$
- $p(x) + Y(p(x)) = Y$
- $p(x)(1 + Y) = Y$
- $p(x) = \frac{Y}{1+Y}$

The equation of the sigmoid function is:

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Logistic Regression

- The student dataset has entrance mark based on the historic data of those who are selected or not selected.
- Based on the logistic regression, the values of the learnt parameters are $\beta_0 = 1$ and $\beta_1 = 8$.
- Assuming marks of $x = 60$, compute the resultant class.

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Logistic Regression

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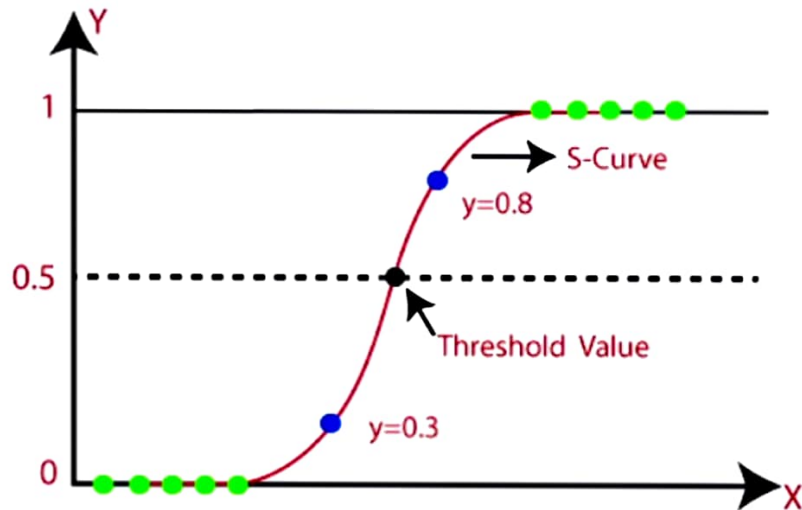
Assume the model suggested by the optimizer for odds of passing the course is,

$$\log(odds) = -64 + 2 * hours$$

Logistic Regression

- We use Sigmoid Function in logistic regression

- $s(x) = \frac{1}{1+e^{-x}}$



Logistic Regression

1. Calculate the probability of pass for the student who studied 33 hours.

$$\bullet \quad p = \frac{1}{1+e^{-z}} \qquad s(x) = \frac{1}{1+e^{-x}}$$

$$\bullet \quad z = -64 + 2 * 33 = -64 + 66 = 2$$

$$\bullet \quad p = \frac{1}{1+e^{-2}} = 0.88$$

• That is, if student studies 33 hours, then there is **88% chance** that the student will pass the exam

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29	0
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28	1
39	1

$$\log(odds) = z = -64 + 2 * hours$$

Logistic Regression

2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

- $p = \frac{1}{1+e^{-z}} = 0.95$
- $0.95 * (1 + e^{-z}) = 1$
- $0.95 * e^{-z} = 1 - 0.95$
- $e^{-z} = \frac{0.05}{0.95} = 0.0526$
- $\ln(e^{-z}) = \ln(0.0526)$

$$\ln(e^x) = x$$

$$-z = \ln(0.0526) = -2.94$$

$$z = 2.94$$

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

Logistic Regression

- $z = 2.94$
- $\log(\text{odds}) = z = -64 + 2 * \text{hours}$
- $2.94 = -64 + 2 * \text{hours}$
- $2 * \text{hours} = 2.94 + 64$
- $2 * \text{hours} = 66.94$
- $\text{hours} = \frac{66.94}{2}$
- **$\text{hours} = 33.47 \text{ Hours}$**

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

- The student should study **at least 33.47 hours**, so that he will pass the exam with more than 95% probability

Logistic Regression

- Linear regression predicts the numerical response but is not suitable for predicting the categorical variables.
- When categorical variables are involved, it is called classification problem.
- Logistic regression is suitable for binary classification problem.
- Here, the output is often a categorical variable.
- The independent variables can be nominal, ordinal, or of interval type.

Logistic Regression – Applications

- Is the mail spam or not spam? The answer is yes or no. Thus, categorical dependent variable is a binary response of yes or no.
- If the student should be admitted or not is based on entrance examination marks. Here, categorical variable response is admitted or not.
- Ecommerce companies can identify buyers if they are likely to purchase a certain product
- Companies can predict whether they will gain or lose money in the next quarter, year, or month based on their current performance

Linear Regression vs Logistic Regression

Linear Regression	Logistic Regression
Used to solve regression problems	Used to solve classification problems
The response variables are continuous in nature	The response variable is categorical in nature
It helps estimate the dependent variable when there is a change in the independent variable	It helps to calculate the possibility of a particular event taking place
It is a straight line	It is an S-curve (S = Sigmoid)